



BACKEND ENGINEERING / CONNECTED SYSTEMS

Alireza Nobakht

Mid-Level ASP.NET Developer | .NET Backend Engineer

E alireza.nobakht2004@gmail.com **M** (+98) 910-977-7936 **L** Tehran, Iran
GH github.com/alirezanobakht2004 **IN** linkedin.com/in/alireza-nobakht **W** Portfolio

PROFESSIONAL PROFILE

Mid-Level ASP.NET Developer and .NET Backend Engineer with 2+ years delivering production ASP.NET Core systems and 3+ years across software development and technical instruction. Builds secure REST APIs, SQL Server data layers, and connected-device solutions using C#, Entity Framework Core, Redis, JWT/RBAC, Clean Architecture, automated testing, secure coding, and CI/CD. Delivered 40% lower average API latency, supported 99.5% reported availability, and produced real-time tooling with zero loss at 115200 baud.

01 TECHNICAL SKILLS

BACKEND & CODING

C#, .NET, ASP.NET Core, Web API, REST APIs, Entity Framework Core, LINQ, SignalR, async/await, OOP, secure coding, coding standards

ARCHITECTURE & SYSTEMS

Applied architectural patterns: Clean Architecture, DDD, CQRS, SOLID, dependency injection, design patterns, Repository / Unit of Work, multi-tenant applications

DATA & PERFORMANCE

Database performance optimization: SQL Server, database design and schemas, stored procedures, indexes, query plans, query optimization, Redis caching, scalability

SECURITY & QUALITY

JWT, OAuth 2.0, ASP.NET Identity, RBAC, FluentValidation, xUnit, NUnit, Moq, unit and integration testing, Swagger / OpenAPI, auditing

DEVOPS & DELIVERY

Docker, CI/CD pipelines, GitHub Actions, Azure DevOps, Git, Linux fundamentals, API versioning, monitoring, production deployment

CONNECTED ENGINEERING

STM32, C/C++, Kotlin / Android, Qt, WPF / WinForms, UART, SPI, I2C, GPS / IMU, multithreading, real-time processing

02 PROFESSIONAL EXPERIENCE

Software Developer & Embedded Systems Engineer

Hiromap (Hirongar Pars Company) | hiromap.com | Tehran, Iran

Feb 2024 - Present

- **Architected** ASP.NET Core services for enterprise ERP and multi-tenant workflows using Clean Architecture, DDD, SOLID, Entity Framework Core, and dependency injection.
- **Delivered** a TypeScript admin dashboard and secure Web APIs spanning **4 operational areas**: users, services, configurations, and RBAC.
- **Reworked** SQL Server schemas, indexes, and queries, reducing average API response time by **40%**.
- **Resolved** routing and performance bottlenecks in a production navigation platform, increasing reported availability to **99.5%**.
- **Created** C++/Qt and C# tools that processed serial streams at **115200 baud** with zero data loss and **40%** lower memory use.
- **Connected 3 platforms** - STM32 firmware, Kotlin/Android, and desktop software - for GPS/IMU processing, motor control, encrypted transfer, and resilient UART/SPI communication.

Technical Instructor & Educational Consultant

Ghalamchi Educational Institute - Tehran, Iran

Aug 2022 - Jun 2023

- **Mentored** students throughout a **10-month engagement** in **3 areas**: programming fundamentals, algorithms, and software-development practices.
- **Prepared** structured lessons, hands-on exercises, and peer-learning sessions across **3 formats** to reinforce programming concepts and collaboration.

Projects, Education & Development

Backend architecture, database performance, security, automated testing, and field-ready connected systems.

ASP.NET DEVELOPER
.NET BACKEND ENGINEER

Tender Management API

Procurement and bidding backend

Jun 2025

Web API EF Core FluentValidation AutoMapper xUnit

- Engineered REST workflows across **4 core domains**: tenders, bids, vendors, and identity, with lifecycle validation and policy-based authorization.
- Standardized **5 cross-cutting concerns**: mapping, validation, dependency injection, exception handling, and OpenAPI documentation.
- Reached **95%+** test coverage and reduced query time by **35%** using xUnit unit and integration testing.

Alibaba Travel Ticket Reservation System

Enterprise multi-modal reservation platform

May 2025

- Designed **4 architecture layers** - Domain, Application, Infrastructure, and Presentation - using Clean Architecture and DDD.
- Modeled booking workflows for **3 transportation modes** with EF Core, SQL Server, stored procedures, indexes, Repository, and Unit of Work.

ASP.NET Core C# SQL Server EF Core Redis Kotlin

- Secured REST APIs for **3 roles** - Admin, User, and Vendor - with JWT, RBAC, validation, standardized errors, and API versioning.
- Cut response time by **60%**, from 450 ms to 180 ms, while targeting **10k+** concurrent users and reporting **99.8%** uptime.

REST
Versioned API
contracts

Redis
Distributed caching

Android
Offline-first sync

Desktop Diagnostics & Cross-Platform Transfer

Real-time tooling and STM32-Android integration

2024 - 2025

- Produced **2 desktop implementations** - C++/Qt and C# - for streaming, filtering, visualizing, recording, and analyzing serial data.
- Supported **3 export formats** - CSV, JSON, and binary - with asynchronous buffering, alerts, and configurable communication parameters.

- Devised encrypted Bluetooth/Wi-Fi transfer with progress, resume, retry, and chunking for files larger than **100 MB**.
- Recorded **99.5%** transfer success and **35%** faster transfers through compression and protocol tuning.

Qt C# WPF / WinForms Kotlin / Android Encryption

Surveying Rover Firmware

Navigation and real-time data collection

Jul 2024

STM32 C/C++ GPS / IMU PID Kalman Filter

- Led STM32 firmware for a **100 Hz** control loop with PID motor control, GPS/IMU fusion, Kalman filtering, and power management.
- Established **2 telemetry paths** - UART and SPI - with fault recovery and error correction for challenging RF conditions.
- Achieved **+/-2 cm** accuracy, **99.9%** transmission reliability over **500 m**, and **8+ hours** runtime.

EDUCATION

B.Sc. Computer Engineering

K.N. Toosi University of Technology
Sep 2022 - Present

Expected graduation: 2026. Teaching Assistant for core programming courses. Study areas include algorithms, databases, software engineering, operating systems, networks, embedded systems, and OOP.

PROFESSIONAL DEVELOPMENT

Azure Fundamentals

In progress
Advanced ASP.NET Core coursework
Clean Architecture & DDD self-study
Microsoft Azure and cloud fundamentals

LANGUAGES & PRACTICES

Persian - Native

English - Intermediate / Professional
Technical communication through instruction and documentation; independent work, cross-functional collaboration, production debugging, and code review.